

ZENITH ORBITAL SYSTEMS

Network Device Management Policy

Effective: March 28, 2026 | Next Review: March 2027 | Version 1.0

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1. Purpose

This Network Device Management Policy establishes a comprehensive, enforceable framework for the secure configuration, monitoring, and lifecycle management of all network devices within Zenith Orbital Systems. Network devices including routers, switches, firewalls, and wireless access points form the critical backbone of ZOS IT infrastructure and the ground station network supporting orbital asset command and control. Their compromise directly threatens the confidentiality, integrity, and availability of every system and mission they support.

This policy implements controls aligned to the SANS/CRF Cybersecurity Policy Template (February 2026), CIS Benchmarks, NIST frameworks, and current industry best practices. By enforcing secure configurations, structured patch management, privileged access controls, and continuous monitoring, this policy seeks to minimize the risk of unauthorized access, data breaches, and network disruptions.

2. Scope

This policy applies to all ZOS employees, contractors, vendors, and third parties who administer, configure, access, monitor, or otherwise interact with network devices owned, leased, or operated by ZOS. It covers:

- Routers and Layer 3 switching equipment
- Layer 2 switches and managed network infrastructure
- Firewalls, next-generation firewalls (NGFW), and UTM appliances
- Wireless access points, WLAN controllers, and wireless infrastructure
- Virtual network devices, software-defined networking (SDN), and SD-WAN components
- Network management, monitoring, and IPAM systems
- Any device that performs network routing, switching, or security filtering functions

3. Principle Statement

Network device security at ZOS is managed on the principle of least privilege and Zero Trust. No device, user, or system is inherently trusted, and every access request to a network device must be continuously verified regardless of origin. This policy operates under the assumption that the network may be compromised at any time, requiring persistent authentication, authorization, and monitoring at every administrative access point.

4. Senior Management Commitment

ZOS leadership is fully committed to the implementation and ongoing enforcement of this Network Device Management Policy. We recognize that our network devices represent critical infrastructure assets whose compromise could result in significant operational, financial, reputational, and legal harm to ZOS and the defense and commercial clients we serve.

Signed: _____ **Date:** _____
Chief Information Security Officer / Senior IT Leadership, Zenith Orbital Systems

5. Roles and Responsibilities

Role	Primary Responsibilities
IT Security Manager	Policy owner; annual review; exception approvals; violation escalation; compliance reporting to leadership
Network Administrator	Implements all NDM controls; maintains device inventory; performs configuration management, patching, and change management
IT Manager	Ensures resource allocation; approves infrastructure changes; reviews exceptions with Security Manager
Security Analyst	Monitors NMS alerts; reviews netflow and log reports; conducts compliance assessments; escalates anomalies
All Staff and Contractors	Must not connect unauthorized devices; must immediately report suspected violations to IT Security Manager

6. Policy Safeguards

6.1 Inventory and Configuration Management

Control ID	Requirement
NDM-01	Maintain a complete, current inventory of all approved ZOS network devices, including make, model, firmware version, physical location, and assigned administrator.
NDM-02	Maintain documented cybersecurity configuration benchmarks aligned to the current CIS Benchmarks for

NDM-13	all authorized ZOS network device types. Benchmarks must be updated within 30 days of any material CIS Benchmark update.
	Use the network management system to perform regular automated comparisons of each device's live configuration against its approved baseline, logging and alerting on any detected changes. Configuration drift alerts must be investigated within 24 hours.

6.2 Secure Management Access

Control ID	Requirement
NDM-03	Manage all ZOS network devices exclusively from a dedicated, approved out-of-band management network subnet, isolated from the production network.
NDM-04	Prohibit management of network devices from any remote or Internet-based network without explicit, time-limited exception approval from the IT Security Manager.
NDM-05	Require that all network device management access is conducted through the approved Privileged Account Management (PAM) system (CyberArk) or dedicated management jump box with full session recording.
NDM-06	Require multi-factor authentication for all administrative access to network devices. Single-factor authentication is not permitted for any device management interface.
NDM-07	Prohibit all unencrypted management protocols. Telnet, HTTP, and SNMPv1/v2c are explicitly banned from all ZOS network devices. Only SSH v2, HTTPS, and SNMPv3 with authentication and encryption are permitted.

6.3 Monitoring and Logging

Control ID	Requirement
NDM-08	Deploy a centralized Network Management System (NMS) or SIEM integration capable of receiving syslog, SNMP traps, and configuration change alerts from all in-scope network devices.

NDM-09	Conduct automated vulnerability scans against all network devices at minimum monthly, using authenticated scanning where possible.
NDM-10	Configure real-time alerting for all critical events including: failed authentication attempts, configuration changes, device unreachability, interface flaps, and CPU or memory threshold breaches.
NDM-11	Retain all network device logs for a minimum of 90 days in the SIEM (Splunk) and 12 months in cold storage. Log integrity must be protected against modification.
NDM-12	Configure all ZOS network devices to export netflow data to the NMS or Splunk SIEM. Retain netflow records for a minimum of 90 days. Netflow analysis enables detection of anomalous traffic, lateral movement, and command-and-control communications.

6.4 Firmware and Lifecycle Management

Control ID	Requirement
NDM-14	Ensure all ZOS network devices operate on the latest vendor-supported firmware version. Patch critical and high severity vulnerabilities within 30 days and medium severity within 90 days. End-of-life devices must be decommissioned per an approved timeline.
NDM-15	All firmware updates must go through the ZOS change management process before deployment to production devices. Emergency patching procedures are pre-approved for CISA KEV-listed vulnerabilities.

6.5 Change Management

All changes to ZOS network device configurations must follow the formal change management process: documented change request, peer review by a second Network Administrator, IT Security Manager approval for high-risk changes, tested in a non-production environment where feasible, implemented during approved maintenance windows, and verified post-implementation with a baseline comparison scan.

7. Legal and Regulatory Obligations

Framework	Relevance
NIST CSF 2.0	Protect (PR) and Detect (DE) functions: secure configuration, access control, continuous monitoring
NIST SP 800-53 Rev 5	SC (System and Communications Protection), CM (Configuration Management), AU (Audit and Accountability)
NIST SP 800-207 Zero Trust Architecture	Mandates continuous verification for all device access; underpins NDM-03 through NDM-07
CIS Controls v8	Controls 1, 4, 5, 8, 12: asset inventory, secure configuration, account management, audit logs, network infrastructure
ISO/IEC 27001:2022	Annex A: Network security (A.8.20), Logging (A.8.15), Secure administration (A.8.31), Change management (A.8.32)
DISA Network Device STIGs	Federal baseline hardening requirements for routers, switches, and firewalls; aligns with NDM-02 and NDM-07

8. Continual Improvement and Sources

This policy is reviewed at minimum annually. Earlier review is triggered by material changes to CIS Benchmarks, NIST guidance, or the ZOS network architecture; a security incident involving a network device; or significant new vulnerabilities affecting ZOS device types.

- SANS/CRF Network Device Management Policy Template (Feb 2026). <https://www.sans.org/information-security-policy/network-device-management-policy>
- CIS Controls v8 and Benchmarks. <https://www.cisecurity.org>
- NIST SP 800-53 Rev 5, SP 800-207, SP 800-128. <https://csrc.nist.gov>
- NSA/CISA Zero Trust Implementation Guideline Primer (Jan 2026)
- CISA Zero Trust Maturity Model v2. <https://www.cisa.gov/zero-trust-maturity-model>
- DISA Network Device STIGs. <https://public.cyber.mil/stigs/>